

Akash Gogate

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Education

University of Wisconsin-Madison

Graduating May 2028

Bachelor of Science in Computer Science — Dean's List Fall 2024, Spring 2025, Fall 2025 — 3.8 GPA

Relevant Coursework: Advanced Algorithms, Data Structures, Machine Organization & Programming, Intro to Computer Engineering, Cryptography, Artificial Intelligence, Computer Science Capstone

Technical Skills

Languages: Java, Python, Rust, C++, SQL, Linux, Bash

Technologies: Kafka, Kubernetes, Docker, MongoDB, PostgreSQL, Redis, FastAPI, REST APIs, Git

Experience

Leidos

May 2025 – Present

Software Engineer Intern

- Eliminated **200+** satellite scheduling conflicts via **Python** memoization & DP across **1,000+** active satellites.
- Built **Kafka+MongoDB** pipeline at thousands of events/sec with dead-letter queue recovery.
- Built **Neo4j** knowledge graph with LLM plugin for **30-engineer** onboarding: **300–500** nodes, **1,000–2,500** edges.
- Built offline **whisper.cpp** ASR for **Android ATAK** at **100%** CoT classification & **~5%** WER.
- Built **DuckDB+Redis** aircraft readiness UI (**React**) with LLM natural language interface; sub-second query latency.
- Automated regression testing via **Docker** & **GitHub Actions** across **5** system components.
- Rewrote satellite scheduling engine in **Rust** for production, leveraging memory safety & async concurrency.

The Kendzierski Lab at the University of Wisconsin-Madison

September 2025 – Present

Student Research Intern

- Reduced scRNA-seq pipeline runtime **80→27 min** via **PyTorch** DataLoader optimization across **4** architectures.
- Cut manual literature review **~60%** via NLP entity extraction across **500+** papers.
- Automated experiment reproducibility via deterministic seeds & versioned checkpoints.

Inspirit AI Research Internship

Sep 2022 – Mar 2023

Machine Learning Research Intern

- Benchmarked **5** classifiers via **scikit-learn** on **70K** dermoscopic images using **OpenCV** preprocessing.
- Achieved **95%** skin cancer accuracy via **Decision Tree**, selected from **5** classifiers by F1-score.

Projects

Lotus Health | [Github Link](#)

March 2026

- Built ICD-10 risk engine on 1,080-node graph over 45M records via SQL RAG with **Cytoscape.js** graph visualization.
- Integrated **Deepgram** voice pipeline for spoken-query ICD-10 risk scoring over **45M** records.
- Designed hybrid **Groq Llama** / Python ICD-10 scoring architecture for reproducible & auditable clinical outputs.

Self-Improving LLM Agent | [Github Link](#)

Jan 2026 – May 2026

- Architected failure-driven self-improvement loop outperforming ReAct **15pp** on 5,000+ tasks via LLM-as-judge harness.
- Built regression benchmark pipeline via **GitHub Actions** to validate agentic self-improvement across 5,000+ tasks.

LLM-Based SAT Practice Test Generator | [Github Link](#)

Jun 2024

- Built full-stack SAT prep app in **TypeScript** & **FastAPI** with **GPT-4o** adaptive question generation & scoring.
- Designed REST API in **FastAPI** routing adaptive questions by user performance history & difficulty progression.

Ecological Conservation Game | [Github Link](#)

Jan 2024

- Engineered **C++** simulation in **Raylib** at **60+ FPS** with O(N) spatial hashing & GPU instancing.
- Engineered multi-generational **4-species** evolution across **10,000+** agents via stochastic fitness mutation.

Leadership

Princeton Racket Club

May 2024 – Aug 2024

Tennis Coach & Tournament Director

- Directed 8 regional tournaments at 380+ match entries with 100% on-time rate as dual director & head coach.
- Coached 8–10 athletes per class in 3 cohorts, advancing 3 juniors 150+ ranking points & 5 adults 0.5 NTRP.

TEL: Spatial Intelligence Ideathon

July 2026

- Earned **2nd place** among **50+** teams via **ZaiNAr** sub-meter positioning & **AR** for airport wayfinding.
- Pitched dual-product airport platform to **5** judges via **ZaiNAr** precision positioning for airline ops.